

CMDA Jupyter Server

<https://hub.jpl-cmda.org>

User Manual

August, 2023

1. Sign in with your account.

Sign in

Username:

Password:

Sign in

2. Choose your server option and start up your server.

Server Options

☒

CMDA
Default environment for using CMDA Webservices. Use this for topics 1-5

☐

RCMES
Regional Climate Model Evaluation System environment. Use this for topic 6.

Start

Click "Start" to start up your server.

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hub.jpl-cmda.org/user/seungwon/lab/workspaces/auto-Q/tree/shared

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Filter files by name

🔍

📁 / shared /

Name	Last Modified
📁 data	2 hours ago
📁 NASA_Summer_School_2023	an hour ago
📁 notebooks	19 days ago

3. Review the directory structure.

Launcher

shared

📖 Notebook



Python 3
(ipykernel)



R



RStudio [↗]



Shiny [↗]

>_ Console



Python 3
(ipykernel)



R

\$ _ Other



Terminal



Text File



Markdown File



Show
Contextual Help

Simple

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Mem: 616.24 MB

Saving completed

Launcher

hub.jpl-cmda.org/user/seungwon/lab/workspaces/auto-Q/tree/shared/NASA_Summer_School_2023/topic_3.ipynb

File Edit View Run Kernel Tabs Settings Help

Filter files by name

/ shared / NASA_Summer_School_2023 /

Name	Last Modified
Groups	2 hours ago
Topic6_RCMES	19 days ago
topic_1.ipynb	6 days ago
topic_2.ipynb	8 hours ago
topic_3.ipynb	a day ago
topic_4.ipynb	an hour ago
topic_5.ipynb	2 hours ago

4. Open the example notebook of your topic.

Launcher topic_3.ipynb

Python 3 (ipykernel)

Topic 3: Land Water Storage Variability

Question 1: Is there an apparent seasonality in GRACE water storage or GLDAS model soil moisture in heavily managed (agricultural) regions? How does that seasonality temporally align with seasonality in precipitation and near-surface temperature, and align with the agricultural seasons when soil water is used for irrigation?

First lets do all necessary imports and start the CMDA services app.

```
[1]: from cmda_utils import area_avg
      from scipy import signal
      import numpy as np
      import xarray as xr
```

We will examine GRACE water storage in 3 regions: Global, South-central Asia, and the Southwest US.

```
[2]: def subset(ds, lat_min, lat_max, lon_min, lon_max):
      lat_mask = (ds.lat > lat_min) & (ds.lat < lat_max)
      lon_mask = (ds.lon > lon_min) & (ds.lon < lon_max)
      return ds.isel(lat=lat_mask, lon=lon_mask)

      grace_global = xr.open_dataset('/home/jovyan/cmda_data/cmip5/others/nasa_grace_zl_a_200204-202105.nc').zl
      grace_sc_asia = subset(grace_global, 23, 35, 66, 96)
      grace_sw_us = subset(grace_global, 31, 42, 236, 258)
      grace = area_avg(xr.concat([grace_global, grace_sc_asia, grace_sw_us], dim='region')).assign_coords(region=['Global
```

The anomalies for each region are plotted below

Simple 0 \$ 4 Python 3 (ipykernel) | Idle Mem: 616.31 MB Saving completed Mode: Command Ln 1, Col 1 topic_3.ipynb

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Filter files by name

/ shared / NASA_Summer_School_2023 /

Name	Last Modified
Groups	2 hours ago
Topic6_RCMES	19 days ago
topic_1.ipynb	6 days ago
topic_2.ipynb	8 hours ago
topic_3.ipynb	a day ago
topic_4.ipynb	an hour ago
topic_5.ipynb	2 hours ago

5. Run each cell by cell.

It will say “busy” when the job is running.
Wait until it says “idle”.

Click it to run each cell.

topic_3.ipynb

```
[13]: import matplotlib.pyplot as plt
      ts = grace.plot.line(x='time')
      plt.title('GRACE Water Storage')
      ts
```

[13]: [<matplotlib.lines.Line2D at 0x7f8026821100>,
<matplotlib.lines.Line2D at 0x7f80247c0970>,
<matplotlib.lines.Line2D at 0x7f80247be280>]

GRACE Water Storage

region

- Global
- SC Asia
- SW US

time

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Filter files by name

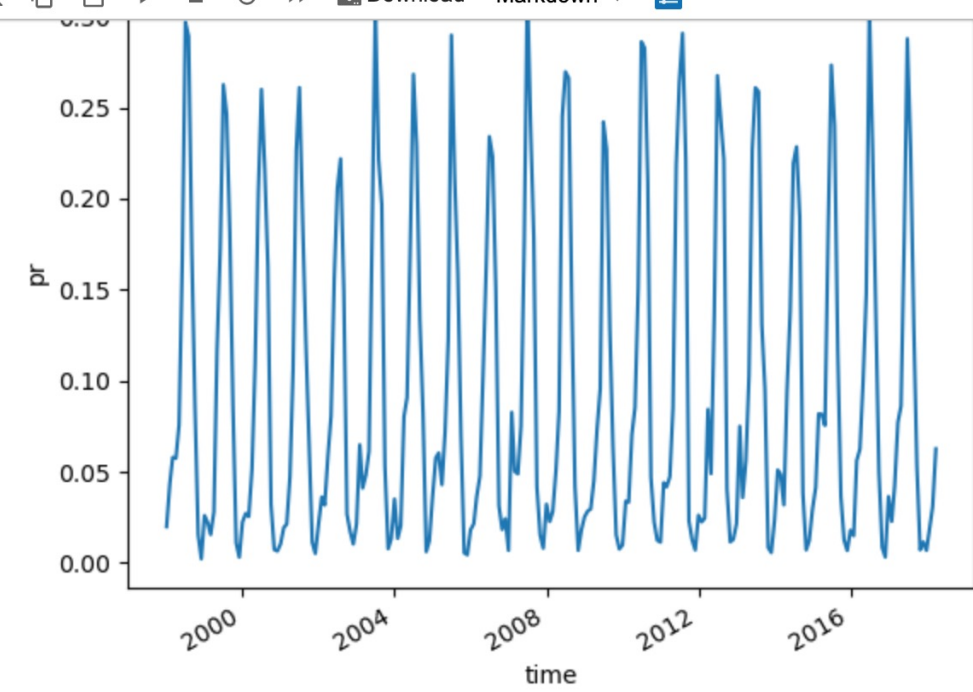
/ shared / NASA_Summer_School_2023 /

Name	Last Modified
Groups	2 hours ago
Topic6_RCMES	19 days ago
topic_1.ipynb	6 days ago
topic_2.ipynb	8 hours ago
topic_3.ipynb	2 minutes ago
topic_4.ipynb	2 hours ago
topic_5.ipynb	2 hours ago

6. Write your own code in a new cell.

Launcher topic_3.ipynb

Python 3 (ipykernel)



Let's make the plots of the time series normalized together and the time values in the common reference point.

```
[ ]: ### Student exercise
```

We can clearly see that water storage is seasonally out-of-phase with precipitation and temperature, with peaks in the former associated with troughs in the latter and vice-versa.

```
[ ]:
```

hub.jpl-cmda.org/user/seungwon/lab/workspaces/auto-Q/tree/shared/NASA_Summer_School_2023/Groups

File Edit View Run Kernel Tabs Settings Help

Filter files by name

/ ... / NASA_Summer_School_2023 / Groups /

Name	Last Modified
Group1	2 hours ago
Group2	2 hours ago
Group3	2 hours ago
Group4	2 hours ago
Group5	2 hours ago
Group6	2 hours ago

6. Share your code and results with your group.

Launcher

topic_3.ipynb

Terminal 1

shared/NASA_Summer_School_2023/Groups

Notebook

Python 3 (ipykernel)

R

RStudio [↗]

Shiny [↗]

Console

Python 3 (ipykernel)

R

Other

Terminal

Text File

Markdown File

Show Contextual Help

Click to open a terminal

Simple 1 \$ 4 Mem: 621.84 MB

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Launcher

hub.jpl-cmda.org/user/seungwon/lab/workspaces/auto-Q/tree/shared/NASA_Summer_School_2023/Groups

File Edit View Run Kernel Tabs Settings Help

Filter files by name

/ ... / NASA_Summer_School_2023 / Groups /

Name	Last Modified
Group1	3 hours ago
Group2	3 hours ago
Group3	3 hours ago
Group4	3 hours ago
Group5	3 hours ago
Group6	3 hours ago

7. Copy your code and results to the group shared folder.

Launcher × topic_3.ipynb × Terminal 1 ×

```
(notebook) jovyan@jupyter-seungwon:~$ ls
cmda_data  shared  work
(notebook) jovyan@jupyter-seungwon:~$ cd shared/NASA_Summer_School_2023/Groups
(notebook) jovyan@jupyter-seungwon:~/shared/NASA_Summer_School_2023/Groups$ ls
Group1 Group2 Group3 Group4 Group5 Group6
(notebook) jovyan@jupyter-seungwon:~/shared/NASA_Summer_School_2023/Groups$
```

Simple 1 4 Mem: 621.84 MB Saving completed Terminal 1